



A PM Model to Deliver a Higher Availability for 794AC Trucks

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

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1.0 Introduction

This Best Practice pioneers building a Preventive Maintenance Model which results in high productivity, quality and safety based on implementing effective and efficient maintenance processes that can be applied to the 794AC Caterpillar electric drive trucks.

In terms of maintenance, customers are demanding:

- 1) Less labor (lower maintenance ratio)
- 2) Less maintenance downtime
- 3) More proactive vs. reactive maintenance on their truck fleets in terms of overall maintenance.

With effective maintenance the customer expects:

- 1) Higher physical availability (performance)
- 2) Quality and reliability
- 3) Lower lifecycle costs.

794AC trucks have recently been introduced into the Brazilian market and so aspects of their maintenance were not well known. The key competition in 320t size class in Brazil is well established and has shown consistent and low variability in the PM downtime; the customers expect similar from the 794AC trucks.

EXAMPLE: '930E-4' KOMATSU (0-5000 hrs)			
PM PERFORMANCE - 2020 (BRAZILIAN CUSTOMER) (14 TRUCKS)			
Type	Average Downtime (Without backlogs) (in hours)	Number of tasks by PM type	% PM Unavailability
250	4,56	123	0,80%
500	13,52	201	2,37%
1000	16,41	237	2,88%
2500	14,48	210	2,54%
5000	17,45	247	3,06%
PA 2020 = 91% and MTBF 2020 = 105 hrs			
OBS: we took off the PM processes wastes (shift changes, delay to wash the trucks, etc)			

PA	91%
UT	87%
PM Hours / Truck Operated Hours	0,039
Truck Operated hrs/ Month	570,0
PM Hours/Month	22,49
PM Unavailability Impact (2020)	3,12%

Competitor PM Unavailability Distribution delivered in Brazil. They put standardization into the PM process downtime to control the wastes and let more time to backlogs.

Starting from the demands as part of the Brazil 794AC NMI, the Sotreq Technical Team utilized the LEAN concept to develop a PM model that would eliminate waste during task execution allowing for more time for backlog execution as required by planning and condition monitoring while maintaining quality and reliability. Basically, what we did was make sure each mechanic had and did **standard work based on previous technical analysis**.

This Best Practice is what Sotreq utilized in terms of maintenance to introduce the 794AC in Brazil and provide the customer with tangible results (increased productivity, quality, reliability, lower direct costs) and practices (lower maintenance ratio). This Best Practice could easily be applied to other dealers and customers introducing new truck fleets into their operations.

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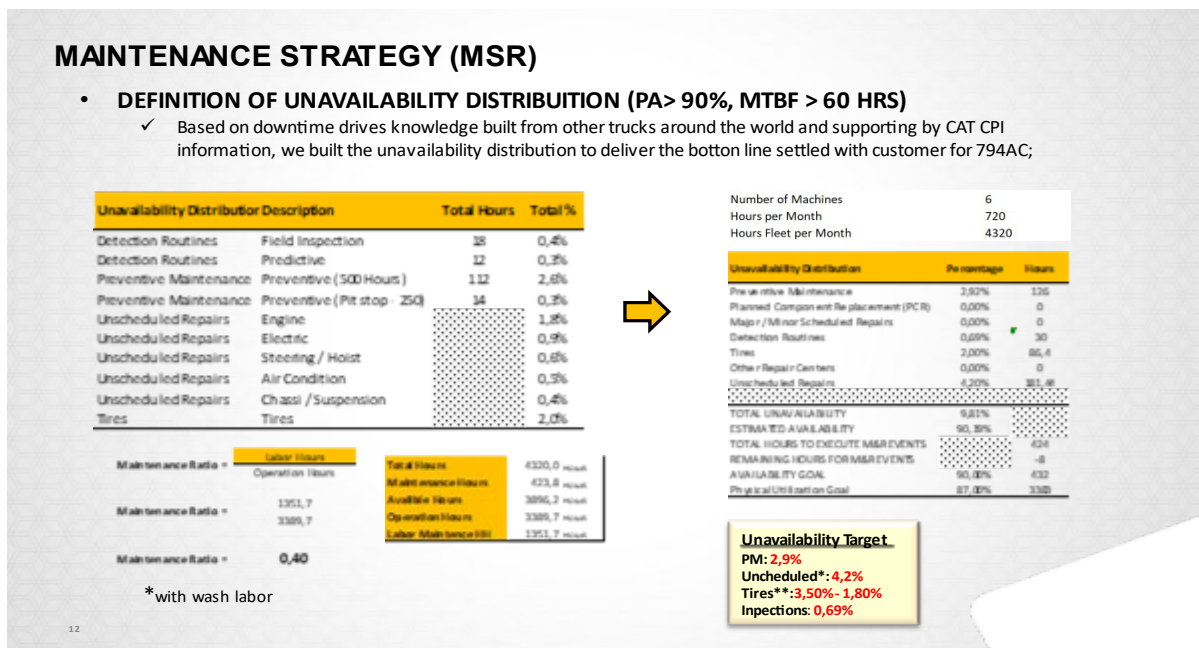
2.0 Best Practice Description

Normally, when developing a “Preventive Maintenance” program for mobile equipment the concentration is on the processes of planning, scheduling, condition monitoring tasks and quality worksheets that will be utilized by the PM team rather than the execution itself.

Rarely, the focus is on identifying models and methods that answer the following:

- How can we achieve the specific goals of PM unavailability (and the targeted performance)?
- What does the maintenance team need to do to achieve those specific goals?
- How should they perform the specific tasks?
- How are they trained and follow instructions for the specific tasks?
- How well do they perform the specific tasks?

Sotreq did extensive research on fleets worldwide to understand the “unavailability of 794AC fleets” and then used that information to design a maintenance strategy to successfully introduce the 794AC’s into Brazil. Sotreq determined that PM duration should be 2.9% of unavailability to achieve a Physical Availability of >90%. As a note, the unavailability due to tires is ~3.5%, which is considered normal for the site of this project with heavy rainfall and poor roads.



Unavailability Distribution Required for Physical Availability <90%

After further research, Sotreq developed a Preventive Maintenance model focused on optimizing their work through a detailed PM plan with all resources (parts, tools, devices, etc.) and workflows (tasks, routes, durations, etc.) well defined and available.

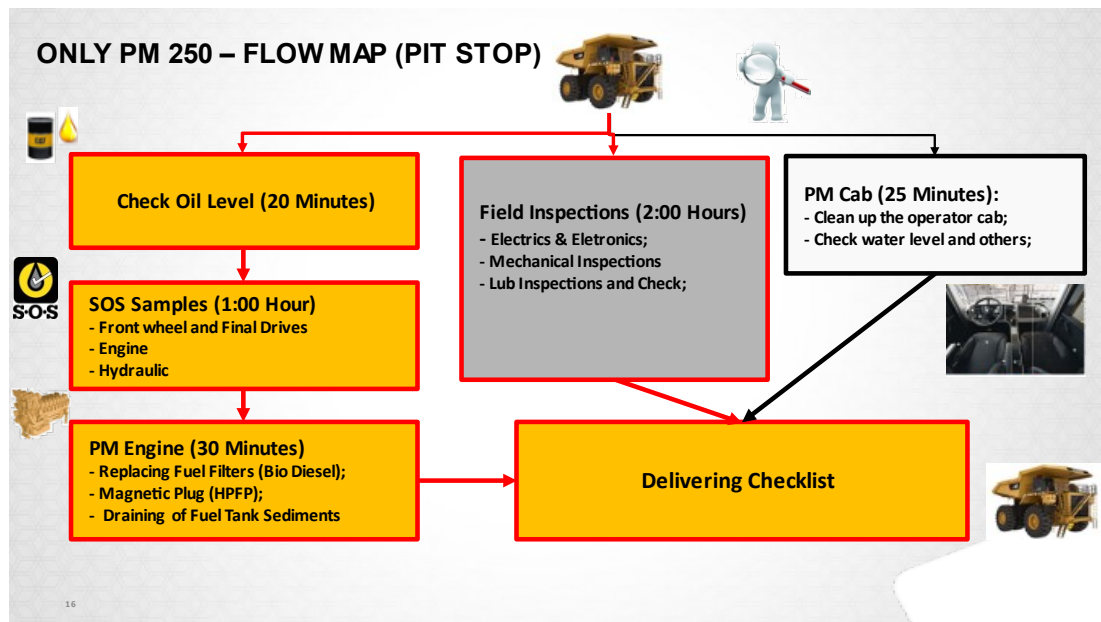
Early in the model development, it was important to establish the maintenance intervals (every 250 or 500 hours) and the workflows of the maintenance execution for each interval (see table below).

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- Mechanical and electrical inspections are done every 125 hours.
- Sotreq decided to have “pit stop” (less than 4 hours, every 250 hours during a “window of opportunity”) for inspections (mechanical, electrical, lubrication) and fuel filter change (needed due to biodiesel saturation). As a note, regular backlogs are not performed during a “pit stop”, only critical ones such as SOS, alarms level 3, etc.
- The PM 500 (500 hour) is the first pass at the structured maintenance execution with an effective distribution of processes, people, and resources to minimize waste during the execution of the maintenance.
- Every 500 hours, this macro flow will be repeated and broken down into the individual activities as shown in the Implementation section below.

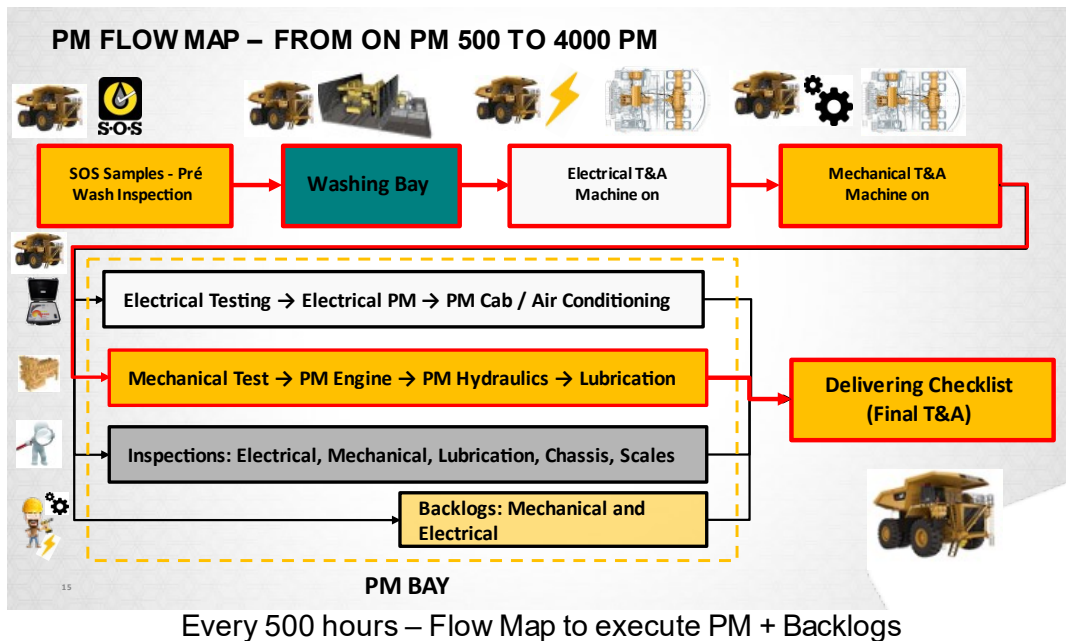
Service	Frequency	Where	125 Hours	250 Hours	125 Hours	500 Hours	125 Hours	250 Hours	125 Hours	1000 Hours	125 Hours	250 Hours	125 Hours	1500 Hours	125 Hours	250 Hours	125 Hours	2000 Hours
Field Inspection: - Mechanical - Eletrical	125 Hours / Weekly	Field / Workshop																
Lubrication Inspection	250 Hours	Field / Workshop																
Pit stop (250 Hours)	250 Hours	Field																
Preventive (500 – 4000 Hours)	500 Hours	Workshop																
Predictive	500 Hours	Field																
Body Inspection	2000 Hours	Workshop																
Body Wear Measurement	Annual																	

Maintenance Intervals for the 794AC



PM 250 hours – Window Opportunity to replace filters and make regular inspections

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3.0 Implementation Steps

First Key Step: Each PM interval was organized to technically balance the activities/tasks without over burdening any one interval knowing that the overall PM unavailability had to be <2.9%. To do this, Sotreq started by evaluating all the PM tasks/activities and determining the duration for them. By adding up all the tasks/activities, the final downtime for each PM interval was known, the maintenance tasks/activities could be distributed by person for efficiency, like in a factory, according to Sotreq’s LEAN model:

- Normally the 2000-hour, 3000 hours & 4000-hour PM’s have extensive maintenance requirements; with this model the maintenance is distributed throughout the other PM’s.
- This distribution is different than what would be seen in a typical “8 Step PM”.
- Some examples include valve recession which was done at 4,000 hours, and has now been moved to the 2500-hour PM and front/rear suspension charging from 1,000 hours to 500-hour PM.

PM Downtime Schematic (not included wash bay + 2 hours)

At this site, Sotreq uses the “maintenance ratio” of 3 PM mechanics + 2 electricians for rapid, systematic LEAN PM execution. As we finish up the PM downtime, the planning team can set up extra time with the same manpower to work on conditional maintenance (backlogs). Below is an example of the time distribution by mechanic for each type of PM.

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DISTRIBUTION BY PM TYPE AND TECHNICIANS

Preventive	Sequenced			Parallel activities						Sequenced	Total PM
	Before Washing	Washing	Mec And Elec Tests	ELET A	ELET B	MEC A	MEC B	MEC C	INSP	Release Check	
500	00:40	02:00	01:35	05:50:00	05:45:00	07:20:00	06:00:00	04:30:00	03:00:00	00:20:00	11:55:00
1000	00:40	02:00	02:15	05:05:00	05:10:00	04:45:00	04:00:00	05:30:00	03:00:00	00:20:00	10:45:00
1500	00:40	02:00	02:05	04:50:00	04:45:00	04:40:00	02:30:00	05:05:00	03:00:00	00:20:00	10:10:00
2000	00:40	02:00	02:15	06:05:00	06:45:00	07:15:00	07:00:00	07:30:00	03:00:00	00:20:00	12:45:00
2500	00:40	02:00	01:35	06:35:00	06:30:00	09:50:00	08:30:00	04:30:00	03:00:00	00:20:00	14:25:00
3000	00:40	02:00	02:15	05:05:00	05:10:00	04:45:00	04:10:00	05:30:00	03:00:00	00:20:00	10:45:00
3500	00:40	02:00	01:35	05:35:00	04:45:00	06:05:00	02:30:00	04:35:00	03:00:00	00:20:00	10:40:00
4000	00:40	02:00	02:15	06:05:00	06:45:00	07:15:00	07:00:00	07:30:00	03:00:00	00:20:00	12:45:00
50F	x	02:00	00:00	00:00:00	00:00:00	04:20:00	04:05:00	03:00:00	02:30:00	00:20:00	06:40:00
250F	x	02:00	00:00	00:20:00	00:00:00	04:40:00	06:05:00	00:30:00	02:00:00	00:20:00	08:25:00
Pit-Stop	x	00:00	00:00	00:20:00	00:00:00	00:40:00	01:50:00	00:30:00	02:00:00	00:20:00	02:00:00

Labor Required

Preventive	Electric	Mechanic
500	11:35:00	17:50:00
1000	10:15:00	14:15:00
1500	9:35:00	12:15:00
2000	12:50:00	21:45:00
2500	13:05:00	22:50:00
3000	10:15:00	14:25:00
3500	10:20:00	13:10:00
4000	12:50:00	21:45:00
50F	0:00:00	11:25:00
250F	1:05:00	11:15:00
Pit-stop	1:05:00	3:00:00

Labor hour for Backlog

Preventive	Electric	Mechanic
500	8:25:00	6:40:00
1000	9:45:00	11:15:00
1500	10:25:00	12:50:00
2000	7:10:00	5:45:00

Preventive	Electric	Mechanic
2500	6:55:00	1:40:00
3000	9:45:00	11:05:00
3500	9:40:00	11:25:00
4000	7:10:00	5:45:00

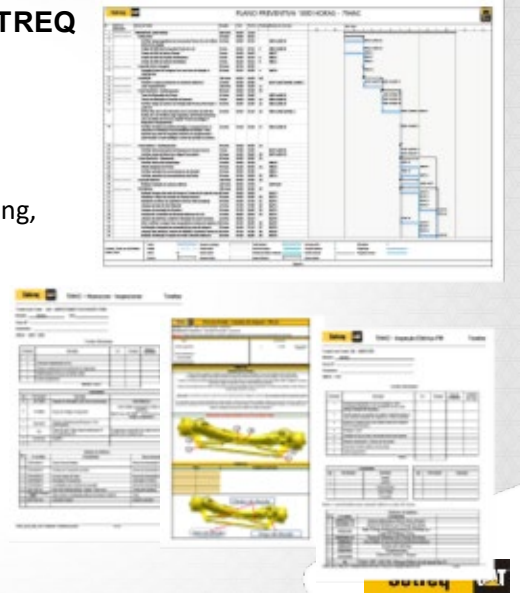
The procedures and checklist for dealing with work safety were developed in order to standardize activities, transforming them into PRO's and reducing daily safety documents.

*PRO's = guidelines.

Second Key Step: To support the execution of the PM intervals in the determined duration, the following were developed and used for each PM interval, for each PM team member, for each task/activity: Gantt charts, playbooks, guidelines, checklists and more Team members were trained by the maintenance supervisor using this same documentation.

PM DOCUMENTATION DEVELOPED BY SOTREQ

- ✓ GANTT Chart BY PM;
- ✓ PM Guidelines by Type (Tasks, Safety, Tooling, Parts, etc);
- ✓ Checklist for all activities;
- ✓ Critical Path Guidelines;
- ✓ "T&A" checklists".



Examples of Gantt Chart for each PM and Checklist Developed

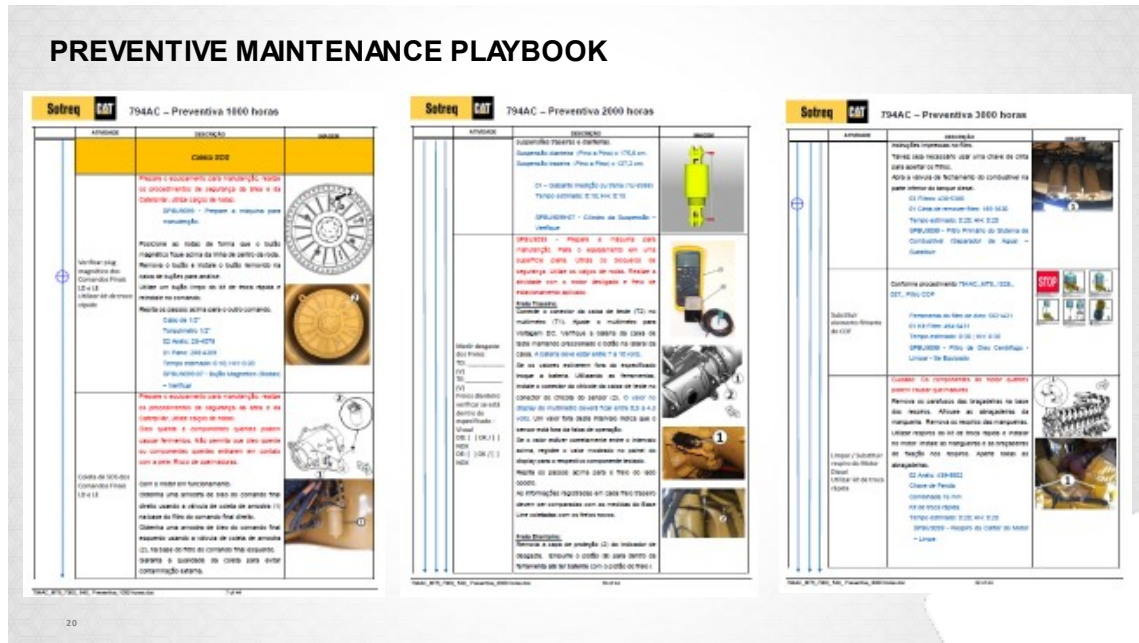
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Examples of Playbooks and Guidelines for each type of PM

It is important to highlight that this technical documentation step for supporting the strategy of the maintenance model created also serves to evaluate and determine what additional technical resources are necessary to execute the specific maintenance activities properly and efficiently.

For example, in terms of velocity, Sotreq analyzed shops from around the world and applied some of the best practices related to shop resources found in this project.

PM STRATEGY (MSR)

- DEVICES AND BEST PRACTICES**
 - ✓ We analyzed for 6 months the Best Workshop practices, lubrication that could be applied in the workshop, without sophisticated process;
 - ✓ We developed devices that made it possible access and consultation of technical material in the area (Electrical, Mechanical, SIS, ...) schemes
 - ✓ We developed devices that enable execution preventive activities of most effective way (Quick plug change kit measure suspension tool)
 - ✓ We standardized the flow previously described with management framework in view of maintenance preventive.

Scheme Cart and Manual Consultation
Screen touch screen 23.8"

Scan Tool Quick height of Suspension

Management routine of the Preventive

Quick change kit Magnetic

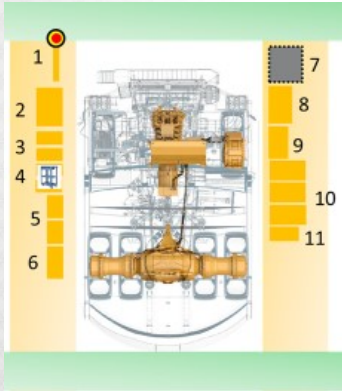
Examples of devices implemented to provide optimized execution thru PM tasks

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Third Key Step: Build an organized PM bay with a layout for the resources (tools, parts, devices, etc.) that would support an efficient workflow and eliminate extra movement of the team members supporting the LEAN concept in our model.

PM LEAN PROCESSES

- PM BAY**
 - ✓ All resources available at the Bay at the start of the PM;
 - ✓ Resources laid out for efficiency of designed PM.



- 1- Management framework in sight
*Pending signal lamp.
- 2- Scheme and Computer Workbench;
- 3- SOS Collection Cart;
- 4- MEC's affection (x03);
- 5- PM Tool Boxes (x02);
- 6- Cabinet PM test tool
- 7- Space for PM material
- 8- Workbench
- 9- Filter Cutting Bench
- 10- Oil Collectors 500 L(x03)
- 11- Low Oil Collector



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Example of PM Bay Implemented in Brazil

The routing/movement for each team member was developed, tested/evaluated, documented, and then implemented. You can see a simulation of the workflow route for each PM team member for the 500 hour and 1000 hours PM's.

PREVENTIVE 500 HOURS – KAMISHIBAI (EXAMPLE)

ELET - A

ELET - B

MEC - A

MEC - B

MEC - C

Preventive: 500 Hours (Step 01)
Total Time: 11:55
HH ELET: 14:45
HH MEC: 19:55

— Electrical Test — Electric PM — PM CABIN	— Mechanical Test — PM Hydraulic — PM Engine — PM Lubrication	
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PREVENTIVE 1000 HOURS – KAMISHIBAI (EXAMPLE)

ELET - A

ELET - B

MEC - A

MEC - B

MEC - C

Preventive: 1000 Hours
Total Time: 10:45
HH ELET: 11:45
HH MEC: 14:40

— Mechanical Test — PM Hydraulic — PM Engine — PM Lubrication	— Electrical Test — Electric PM — PM CAB	
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Example PM 500 and 1000 of flow process for each technician

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Fourth Key Step: Training is important to the success of the maintenance. Each team member was trained by the maintenance supervisor and technical team.

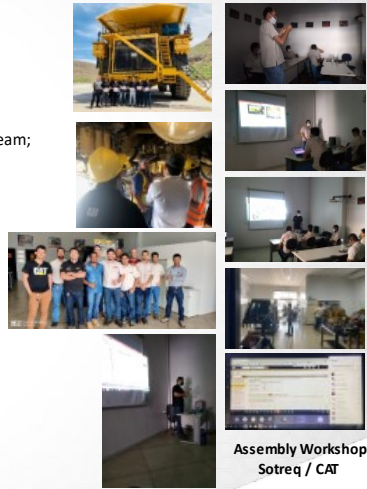
Training consisted of the following:

- Overall knowledge of truck/truck systems
- Specific Tasks/Activities that each team member may be required to perform
- How to safely do the tasks/activities
- How each team member's work ties into the reliability (MTBF) and overall performance of the fleet....amongst more.

Plans are in place to evaluate each team member and the fleet performance on a regular basis and re-train the team as required.

TRAINING AND WORKSHOPS

- **DEVELOPMENT DEEP KNOWLEDGE ABOUT 794AC**
 - ✓ 12 People in Service Training between CAT and SOTREQ (Brazil)
 - ✓ 15 days of interaction between Project Development Team and Execution Team;
 - ✓ More than 100 hours in virtual workshops around the PM Documentation;
- **SPREADING THE TECHNICAL KNOWLEDGE THRU TEAMS**
 - ✓ Engineering settled a robust training grid with CT and SAP ;
 - ✓ + 120 hours of service training
 - ✓ + 120 hours of PM service training
 - ✓ +96 hours of CAT online training and workshop for Support to the SOTREQ
 - ✓ 20 hours of PM Service Training Guideline and M&R routines for PM programmers and Supervisors ;
 - ✓ + 20 hours of workshops on improvements, parts, components, processes, etc. with product support ;



Assembly Workshop
Sotreq / CAT

Sotreq CAT

Training Plan executed during the PM Project

4.0 Benefits

This model supports delivering the following results consistently:

- High Productivity based on achieving Planned PM Unavailability
- Quality & Reliability
- Safety

As you can see in the below graphic, the Sotreq Maintenance Model is delivering better than the planned 2.9% PM unavailability allowing us to meet/exceed targeted Physical Availability. Additionally, Sotreq is delivering great reliability to the 794AC fleet – MTBF > 60 hours even though Sotreq is executing more technical tasks for the 794AC fleet compared to the competition.

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EXAMPLE: '794AC' BRAZIL (0 - 2000 Hours)			
PM PERFORMANCE - 2021 (BRAZILIAN CUSTOMER)			
Type	Average Downtime (Without backlogs) (in hours)	Number of tasks by PM type	% PM Unavailability Month
250 H	3.0	24	0.53%
500 H	12.4	86	2.18%
1000 H	10.1	92	1.78%
1500 H	10.0	90	1.76%
2000 H	12.3	93	2.16%
TOTAL	56.8		

For 2000 Hours cycle is consider 04 x 250 Hours + 500 Hours + 1500 Hours + 2000 Hours

OBS: Some product issues affected the PA, without this matter PA was 93%. MTBFc = 62 hrs

RECOMMENDATION: Can you share the graphs and trends to support the statements provided above?

2000 HOURS CYCLE	
PA	90.80%
UT	87%
PM Hours / Truck Op. Hours	0.028
Truck Operated Hrs. / Month	568.8
PM Hours / Month	16.2
PM Unavailability Impact (2021)	2.8%

5.0 Resources Required

Project Investment Cost – Around \$ 3,000 USD

- Facility – \$500 USD (paint the shop floor)
- Tools – \$500 USD (write board and stands)
- Support equipment – \$1,000 (repair the kidney looping carts)
- Training – \$ 1,500 USD (travel costs, and documentation)
- Information System – \$500 (books and printing)

6.0 Supporting Attachments / References

DMAIC Methodology, 6 Sigma Process, Lean Manufacturing Guidebook, VPS – Customer Production System Guideline, 794AC Service Manual.

7.0 Related Best Practices

We consider as support information some Best Practice:
Unavailability Distribution – Effectively Manage the Unavailability of your Fleet

8.0 Acknowledgements

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